BENGALURU TRAFFIC FLOW, CONGESTION MANAGEMENT & MOBILITY OPTIMIZATION

Traffic Performance & Optimization Report

Executive Summary:

This project examines traffic flow and congestion patterns in Bengaluru using a supplied traffic dataset and a Python analytical workflow. The available project artifacts document data preparation and quality checks, descriptive statistics, area and road/intersection comparisons, relationship analysis, derived screening indicators, and an interactive Streamlit dashboard.

The inspected project report describes 8,936 observations across eight named areas and 16 road/intersection names, spanning 1 January 2022 to 9 August 2024. In that dataset version, mean congestion was 80.82, mean average speed was 39.45, mean travel-time index was 1.376, and mean road-capacity utilization was 92.03%. The source files do not document the units or full definitions of these measures; values are reported on the dataset's own scales.

Koramangala had the highest reported area-level mean congestion (93.99), followed by M.G. Road (90.58) and Indiranagar (87.64); Electronic City had the lowest (54.45). Sony World Junction (94.13) and Sarjapur Road (93.85) had the highest reported road/intersection mean congestion. Traffic volume was positively associated with congestion (Pearson $r \approx +0.837$) and negatively associated with average speed ($r \approx -0.341$). These are descriptive associations, not proof of causation.

The dashboard supports interactive exploration through filters, KPI cards, trend charts, area and road comparisons, scatter plots, and context fields where available. Important limitations include the absence of time-of-day, road type, zone, coordinates, commercial-hub distance, demographic variables, and event start/end timestamps. Therefore, hourly rush-hour analysis, precise mapping, demographic equity conclusions, incident recovery duration, and validated congestion prediction cannot be claimed from these fields. Scenario simulations are illustrative, not predictions of real-world congestion improvement.

1. Project Background

1.1 Problem statement:

Urban congestion can reduce travel efficiency, increase delays, and place pressure on road capacity. Analysis of traffic volume, speed, congestion, and available context can help identify locations for closer investigation and support evidence-informed planning.

1.2 Objectives:

- Prepare and validate the supplied traffic dataset.
- Summarize traffic volume, average speed, congestion, travel-time index, and capacity utilization.
- Compare available area and road/intersection profiles.
- Explore relationships between demand, speed, congestion, and capacity utilization.
- Create derived indicators for screening unusual or high-congestion observations, labelling proxies clearly.
- Present selected results in an interactive Streamlit dashboard.
- Recommend monitoring and follow-up studies while documenting limitations.

1.3 Scope:

This is a descriptive analytics project with rule-based screening indicators and an illustrative scenario. It is not presented as a validated traffic forecasting or causal-impact model.

2. Dataset Description:

The inspected source dataset contains 8,936 rows and 16 original columns: Date, Area Name, Road/Intersection Name, Traffic Volume, Average Speed, Travel Time Index, Congestion Level, Road Capacity Utilization, Incident Reports, Environmental Impact, Public Transport Usage, Traffic Signal Compliance, Parking Usage, Pedestrian and Cyclist Count, Weather Conditions, and Roadwork and Construction Activity.

Attribute	Reported detail
Observations	8,936 records
Area coverage	8 named areas
Road/intersection coverage	16 named locations
Date coverage	1 Jan 2022 – 9 Aug 2024
Unique dates	952
Coverage	4–15 records per date; not a complete fixed panel

3. Data Preparation and Quality Checks

3.1 Preparation steps documented:

- Standardize column names for consistent Python access.
- Parse the date field as datetime.
- Derive date-related fields such as weekday, month, year, and weekend indicator.
- Save a separate cleaned/enriched dataset while preserving the original.
- Create descriptive proxies such as traffic-volume-to-speed delay index, high-congestion and high-capacity flags, volume z-score/anomaly flag, risk flag, rolling speed variability, and baseline deviations.

3.2 Reported validation results:

Check	Reported result / interpretation
Missing values	No missing cells reported in inspected audit
Exact duplicate rows	None reported
Duplicate date-area-road keys	None reported
Negative traffic volumes	None reported
Non-positive speeds	None reported
Date coverage	952 unique dates
Coverage consistency	4–15 records per date; interpret trends with care

4. Analytical Methodology

4.1 Descriptive analysis:

The workflow summarizes and compares congestion, speed, traffic volume, and capacity utilization across available areas and road/intersection categories.

4.2 Correlation analysis:

Pearson correlations describe linear association between numeric measures. They do not prove that one variable causes another.

4.3 Derived indicators and proxy measures:

Indicator	Interpretation and caution
Delay index proxy	Traffic volume divided by average speed; not measured delay in minutes.
High-congestion flag	Threshold-based label; threshold and source scale must be documented.
High-capacity flag	Screening label, not direct proof of road failure.
Volume anomaly flag	Statistical deviation flag; does not prove an accident occurred.
Risk flag	Rule-based screening indicator, not a trained predictive model.
Baseline deviation	Difference from a descriptive baseline, not a causal forecast.

4.4 Temporal limitations:

The inspected source has a date field but no time-of-day field. Daily, monthly, weekday, and weekend summaries may be possible where implemented, but reliable hourly or morning/evening rush-hour analysis is not supported. Observation coverage varies by date.

5. Key Findings

5.1 Overall indicators:

Metric	Reported value	Interpretation
Mean congestion	80.82	**The average congestion level is 80.82 out of 100, indicating a high level of traffic congestion across the recorded observations.**
Mean average speed	39.45	**The average recorded traffic speed is 39.45, showing the overall average speed across the observed roads and locations.**
Mean travel-time index	1.376	**The average travel-time index is 1.376, indicating a higher travel-time index across the recorded observations.**
Mean road-capacity utilization	92.03%	**The average road-capacity utilization is 92.03%, indicating that the recorded roads are operating at a high level of their available capacity.**

5.2 Area comparison:

Area	Mean congestion
Koramangala	93.99
M.G. Road	90.58
Indiranagar	87.64
Electronic City	54.45

Koramangala had the highest reported area-level mean congestion and Electronic City the lowest. These are useful screening comparisons but do not explain the causes or prove which intervention will work.

5.3 Road/intersection comparison:

Road/intersection	Mean congestion
Sony World Junction	94.13
Sarjapur Road	93.85

These locations are candidates for further field assessment, signal-operation review, and more granular traffic collection.

5.4 Relationships between measures:

Variable pair	Reported Pearson correlation	Interpretation
Traffic volume and congestion	+0.837	Positive association in inspected data.
Traffic volume and average speed	−0.341	Negative association in inspected data.
Capacity utilization and congestion	+0.865	Positive association in inspected data.

These correlations are descriptive and may reflect coverage, field definitions, or other unmeasured factors. They are not proof that a specific increase in volume causes a quantified congestion increase.

6. Interactive Streamlit Dashboard

The project includes a Streamlit dashboard built with Pandas and Plotly. The inspected dashboard code contains KPI cards and interactive charting intended to make the analysis easier to explore.

6.1 Dashboard components documented in code:

- File upload and local-file loading options.
- Sidebar filters for available area, road/intersection, and date range.
- KPI cards for records, mean congestion, mean speed, and mean traffic volume.
- Selectable weekly trend chart.
- Average congestion by area.
- Comparison of the most congested road/intersection locations.
- Traffic-volume-versus-congestion and speed-versus-congestion scatter plots.
- Weather and roadwork context tables when fields are available.
- Table of high-congestion records.

6.2 KPI interpretation:

The dashboard's KPI cards summarize the filtered records. 'Total Records' counts rows after filtering; average congestion, average speed, and average traffic volume are means over filtered records. These cards do not implement all four KPIs in the trainer's reference solution (average speed, peak load, delay index, and reliability). Peak load requires a valid peak-period indicator; the inspected dataset lacks time-of-day.

6.3 Dashboard limitations:

Dashboard results depend on the dataset loaded and selected filters. Before submission, verify the latest script, CSV path, column-cleaning logic, and successful app execution. Scatter plots can look dense when records overlap; sampling and marker opacity reduce but do not eliminate this.

7. Risk Screening, Bottlenecks, and Scenario Analysis

7.1 Bottleneck screening:

Area and road/intersection rankings identify locations with higher average congestion. A high average is a screening signal, not a definitive explanation of a bottleneck. Confirmation requires granular flow data, road geometry, signal timing, incident details, and field observations.

7.2 Risk indicators:

The enriched dataset includes risk and anomaly-related variables. These are rule-based or statistical screening indicators, not a validated predictive system unless a model has been trained and evaluated on held-out data.

7.3 Scenario simulation:

The project includes an illustrative traffic-volume-reduction scenario. Selected records have their volume values arithmetically reduced by a chosen percentage and scenario summaries are compared with baseline values. This does not establish how congestion or travel time would respond. A validated traffic-response model or intervention data would be needed to estimate real operational impact.

8. Structural Weaknesses and Evidence:

Potential issue	Evidence available	Still needed
High-average-congestion locations	Area and road/intersection rankings	Field checks, signal timing, road geometry, granular observations.
Demand and capacity pressure	Capacity-utilization and congestion association	Confirm metric definitions and capacity measurement.
Unusual observations	Volume z-scores/anomaly flags and baseline deviations	Incident logs and event timestamps to establish causes.
Uneven data coverage	4–15 records per date reported	Consistent repeated measurements by location and period.
Limited temporal diagnosis	Date present; time-of-day absent	Timestamped or hourly observations.
Limited spatial/network diagnosis	No coordinates, road type, zone, or origin-destination data	GIS coordinates, road network, road classes, travel paths.

9. Recommendations and Improvement Roadmap:

Priority	Recommended action	Suggested monitoring
Validate hotspots	Investigate high-average-congestion locations before committing to interventions.	Repeated observations, queue length, speed, congestion by period.
Review signal operations	Assess signal coordination and junction timing where field evidence supports it.	Waiting time, queue length, throughput, before/after comparisons.
Public transport integration	Assess route connectivity, stop access, and service capacity in relevant corridors.	Ridership, service reliability, mode-share measures if available.
Curbside activity	Review parking, loading, and roadside activity at validated bottlenecks.	Blocked-lane frequency, parking occupancy, corridor speed.
Smart monitoring	Collect consistent traffic counts and timestamped incident/event logs.	Data completeness, update frequency, incident response time.
Evaluate interventions	Use a documented baseline and comparable post-intervention period.	Speed, volume, congestion, travel-time index, uncertainty.

Road widening, congestion charging, or other major interventions should not be recommended solely from descriptive averages. They require feasibility analysis, environmental and equity review, local consultation, and evaluation of alternatives.

10. Monitoring and Evaluation Framework:

Frequency	Monitoring focus	Purpose
Daily / operational	Volume, speed, congestion, incidents, data completeness where collected	Identify immediate anomalies and operational issues.
Weekly	Hotspot rankings and short-term trend comparisons	Track recurring patterns and prioritize review.
Monthly	Area/road profiles, capacity indicators, risk flags	Assess persistence and investigate changes.
Quarterly	Intervention reviews and data-quality assessment	Compare outcomes against baselines.
Yearly	Infrastructure performance and longer-term trends	Support planning while accounting for coverage changes.

For credible intervention evaluation, retain a baseline period, record intervention dates and locations, compare like-for-like periods, document other changes, and avoid attributing differences to an intervention without a suitable evaluation design.

11. Data Limitations and Responsible Interpretation:

- Units and some field definitions are not documented in inspected materials.
- No time-of-day field is available; hourly/rush-hour claims cannot be verified.
- Road type, zone, coordinates, commercial-hub distance, and demographic fields are absent.
- Incident start/end timestamps and verified recovery labels are unavailable.
- Uneven record counts by date may affect comparisons and trends.
- Correlation does not establish causation.
- The risk flag is a screening rule, not a validated predictive model.
- The volume-reduction scenario is illustrative and does not predict congestion improvement.
- Recheck final values against the exact final CSV and notebook before publication.

12 . Conclusion:

This project demonstrates a practical traffic analytics workflow using Python, Pandas, visualization, derived indicators, and a Streamlit dashboard. The inspected results identify locations with relatively high average congestion and descriptive associations between traffic volume, speed, congestion, and capacity utilization. These findings can guide further investigation, but reliable operational decisions require confirmed data definitions, richer temporal and spatial coverage, field validation, and measured evaluation of interventions.

Next steps are to validate the final notebook and dashboard, confirm reported figures against the final cleaned/enriched dataset, collect timestamped and spatial traffic data, and evaluate interventions against a clear baseline. Scenario and risk outputs should remain labelled as illustrative or screening-based until validated.